

Homologous series

Learning outcomes

At the end of this section you should be able to:

- (S3.2.3) Identify the following homologous series: alkanes, alkenes, alkynes, halogenoalkanes, alcohols, aldehydes, ketones, carboxylic acids, ethers, amines, amides and esters.
- (S3.2.4) Describe and explain the trend in melting and boiling points of members of a homologous series.

The alkanes are an extremely useful class of organic compounds. They have a variety of uses which depend on their physical properties. Lighter alkanes such as propane and butane make good fuels for cooking because they are gases and can be easily stored. Alkanes such as gasoline, diesel and kerosene are also used as fuels, but these are liquids rather than gases. Being liquids, they can be easily transported. And heavier alkanes, which tend to be solids, are used to make roads (Figure 1).

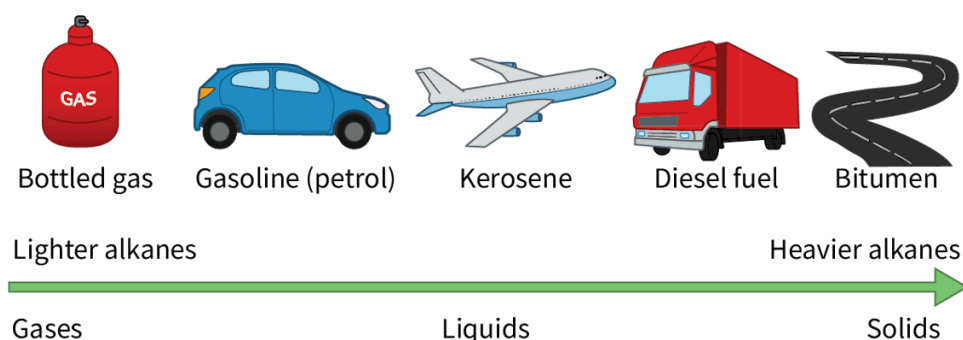


Figure 1. Alkanes have a wide variety of uses that depend on the change in their physical properties.

 More information for figure 1

This image is a diagram illustrating the change in physical states of alkanes, ranging from gases to solids. It includes illustrations of a gas bottle, a car, an airplane, a truck, and a road, representing bottled gas, gasoline (petrol), kerosene, diesel fuel, and bitumen, respectively. Beneath these illustrations is a horizontal arrow labeled "Lighter alkanes" to "Heavier alkanes". Below the arrow, the words "Gases", "Liquids", and "Solids" are positioned under the images of the gas bottle, car and airplane, truck, and road, respectively. This arrangement shows the transition from lighter to heavier alkanes and their corresponding physical states and typical uses.

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You may have noticed that the alkanes mentioned in this section have different physical states that determine their uses. But why do organic compounds such as the alkanes have different physical properties? In this section, you will learn about the factors that affect physical properties of organic compounds such as their melting and boiling points.

In the previous [section S3.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/), you learned that organic compounds are divided into different classes based on the functional group that they contain. Successive members of compounds with the same functional group are known as a homologous series. As an example, let's consider the first four members of the alkane homologous series, methane to butane (**Figure 2**). What do you notice about the successive members of the alkane homologous series? How do they differ? Is there any feature common to each member?

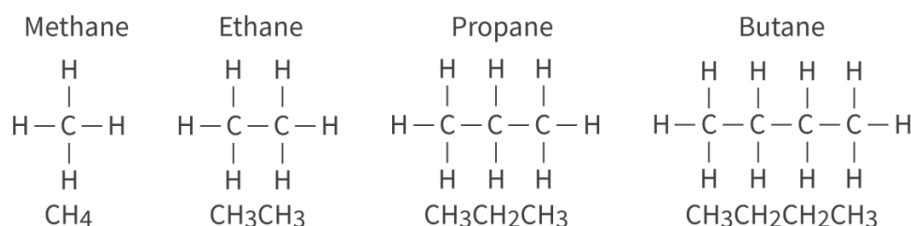


Figure 2. The full structural formulas of the first four members of the alkane homologous series, methane to butane.

 More information for figure 2

This image shows the full structural formulas of the first four members of the alkane homologous series: methane, ethane, propane, and butane. Each structure is depicted vertically with carbon (C) atoms forming a backbone, and hydrogen (H) atoms attached around them. Methane (CH₄) is shown with one carbon atom bonded to four hydrogens. Ethane (C₂H₆) has a two-carbon chain with each carbon bonded to sufficient hydrogens to complete four bonds per carbon. Propane (C₃H₈) continues the chain with three carbon atoms, each bonded to hydrogens. Lastly, butane (C₄H₁₀) extends the chain to four carbon atoms with hydrogen completing each carbon's valency. The diagram visually illustrates how each successive member in the alkane series has an additional CH₂ group compared to its predecessor.

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A look at the full structural formula of each member reveals that they belong to the same class (the alkanes) but they differ by a common structural unit, that is a $-\text{CH}_2-$. Therefore, the molar mass of each member of the series increases by roughly 14 g mol^{-1} (the mass of one carbon atom and two hydrogen atoms). Because of this increase in molar mass, there is also a gradation (a gradual increase) in physical properties such as melting point and boiling point. The graph in **Interactive 1** shows the melting points and boiling points for the first six members of the alkane homologous series.

Interactive 1. A Graph of the Melting and Boiling Points of the First Six Members of the Alkane Homologous Series.

 More information for interactive 1

The interactive displays a bar chart comparing the melting and boiling points of alkanes from methane (CH_4) to hexane (C_6H_{14}). The x-axis represents different alkanes, while the y-axis represents temperature in degrees Celsius. Light blue bars indicate melting points, and red bars indicate boiling points. Melting points are negative for all compounds, decreasing further from methane to butane (C_4H_{10}) before increasing slightly for pentane (C_5H_{12}) and hexane. Boiling points increase progressively from methane to hexane, with methane being the lowest and hexane the highest. The general trend shows that as the carbon chain length increases, both melting and boiling points rise. The approximate data is as follows. Methane has a melting point of negative 182 degrees Celsius and a boiling point of negative 162 degrees Celsius. Ethane has a melting point of negative 183 degrees Celsius and a boiling point of negative 89 degrees Celsius. Propane has a melting point of negative 188 degrees Celsius and a boiling point of negative 42 degrees Celsius. Butane has a melting point of negative 140 degrees Celsius and a boiling point of negative 0.5 degrees Celsius. Pentane has a melting point of negative 130 degrees Celsius and a boiling point of 36 degrees Celsius. Hexane has a melting point of negative 95 degrees Celsius and a boiling point of 69 degrees Celsius. The hotspot is given as a plus sign on each graph, data are as follows. Hot spot 1 for C_4H_{10} , $M = 58.14 \text{ g per mol}$. Hot spot 2 for C_2H_6 , $M = 30.08 \text{ g per mol}$. Hot spot 3 for C_6H_{14} , $M = 86.20 \text{ g per mol}$. Hot spot 4 for C_3H_8 , $M = 44.11 \text{ g per mol}$. Hot spot 5 for C_5H_{12} , $M = 72.17 \text{ g per mol}$. Hot spot 6 for C_4H_4 , $M \text{ equals } 16.05 \text{ g per mol}$.

What do you notice about the trend in the boiling point and melting point as the molar mass increases? Intermolecular forces were covered in [section S2.2.8a](#) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/print/)

[753301/book/intermolecular-forces-id-44359/](#)), which may help you to explain the trend (hint: alkanes are non-polar molecules). Check your answer when you are ready.

Alkanes are non-polar molecules, so the only intermolecular forces between the molecules are London dispersion forces. The strength of the London dispersion forces between the molecules depends on the contact surface area between the molecules. Larger molecules with a higher molar mass have a greater contact surface area and stronger London dispersion forces. So, as the molar mass increases, the strength of the London dispersion forces also increases and so do the melting point and boiling point.

In addition to London dispersion forces, there are two other types of intermolecular force covered in [section S2.2.8a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/intermolecular-forces-id-44359/>), which are dipole–dipole attractions and hydrogen bonds. The type of intermolecular force that exist between the molecules of a compound depend on three basic factors:

1. The molar mass of the compound, which was discussed earlier in this section.
2. The structure of the molecule: whether it is straight-chain or branched.
3. The functional group present in the molecule, which has an effect on the polarity of the molecule.

Next, we will look at the second and third factors in more detail starting with the structure of the compound. Take the compounds pentane and 2,2-dimethylpropane as an example (**Figure 3**). Both compounds have the same molecular formula and molar mass but the boiling points are different. Why do you think this is?

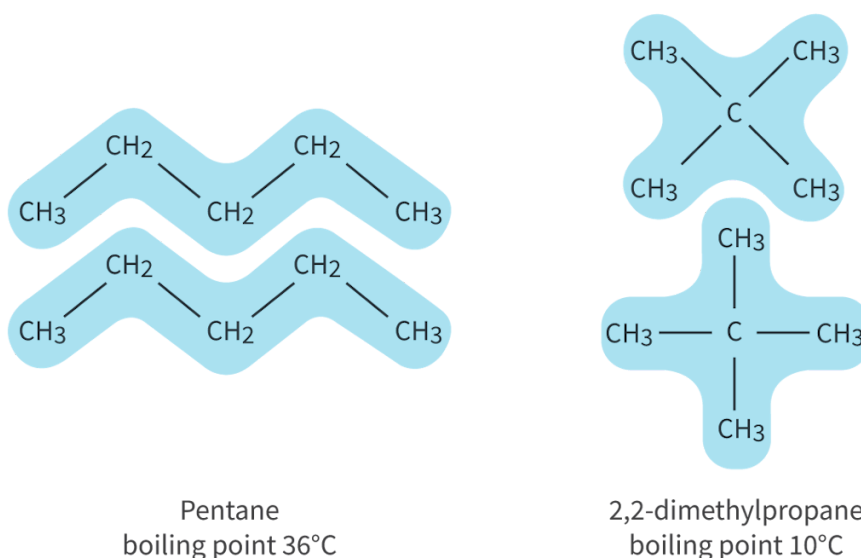


Figure 3. Pentane and 2,2-dimethylpropane have different boiling points despite having the same molar mass.

 More information for figure 3

The image shows structural diagrams of two chemical compounds: pentane and 2,2-dimethylpropane.

On the left, the structure of pentane is depicted. It appears as a zigzag line, representing a straight-chain molecule composed of three CH₃ groups at each end and two CH₂ groups in the middle. The boiling point is indicated as 36°C.

On the right, the structure of 2,2-dimethylpropane is shown. It is depicted as a cross shape, representing a branched-chain molecule with a central carbon (C) atom bonded to four CH₃ groups. The boiling point is noted as 10°C.

Pentane's straight-chain structure allows for more interaction between molecules, resulting in a higher boiling point, whereas the branched structure of 2,2-dimethylpropane leads to less molecular interaction and a lower boiling point.

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A closer look at their structures reveals that pentane is a straight-chain molecule and 2,2-dimethylpropane is a branched-chain molecule. The branches of 2,2-dimethylpropane prevent the molecules from getting close together, which results in less surface area contact between the molecules and weaker London dispersion forces. The straight-chain molecules of pentane can get closer together, which results in a greater surface area contact and stronger London dispersion forces. This explains why the straight-chain molecule has a higher boiling point than the branched-chain molecule.

The third factor that affects the boiling points of organic compounds is the functional group present in the molecule. The presence of different functional groups has an effect on the polarity of the molecule, which determines the type of intermolecular force that act between the molecules. Recall that the strength of the three types of intermolecular force are (from weakest to strongest):

London dispersion forces < Dipole–dipole attractions < Hydrogen bonding

So we can predict that molecules that are able to form hydrogen bonds between their molecules will have higher boiling points and those with only London dispersion forces, lower boiling points. Exactly how the polarity of the molecules determines the type of intermolecular forces that act between the molecules of a compound is as follows.

- Non-polar compounds such as the alkanes, alkenes and alkynes will only have London dispersion forces acting between their

molecules and tend to have lower boiling points, in other words they are more volatile.

- Polar compounds such as aldehydes and ketones have not only London dispersion forces acting between their molecules but also dipole–dipole attractions. This results in these compounds having higher boiling points than non-polar organic compounds of similar molar mass.
- Compounds with either N–H or O–H bonds, such as the alcohols, carboxylic acids and amides, have hydrogen bonds acting between their molecules. Because hydrogen bonds are the strongest type of intermolecular force, these compounds tend to have higher boiling points and are less volatile.

Table 1 shows the molar masses, types of intermolecular forces and the boiling of melting points of different classes of organic compounds. Note that each compound has a similar molar mass so we can assume that the London dispersion forces that act between each molecule are relatively equal in strength.

Table 1. The molar masses, types of intermolecular forces and the boiling points of different classes of organic compounds.

Compound	Class	Molar mass (g mol ⁻¹)	Type of intermolecular force	Boiling point (°C)
Ethanamide	Amide	59.1	Hydrogen bonding Dipole–dipole attractions London dispersion	222

Compound	Class	Molar mass (g mol ⁻¹)	Type of intermolecular force	Boiling point (°C)
			forces	
Ethanoic acid	Carboxylic acid	60.1	Hydrogen bonding Dipole—dipole attractions London dispersion forces	118
Propan-1-ol	Alcohol	60.1	Hydrogen bonding Dipole—dipole attractions London dispersion forces	97
Propanone	Ketone	58.1	Dipole—	56

Compound	Class	Molar mass (g mol ⁻¹)	Type of intermolecular force	Boiling point (°C)
			dipole attractions London dispersion forces	
Propanal	Aldehyde	58.1	Dipole—dipole London dispersion forces	49
Propanamine	Amine	59.1	Hydrogen bonding Dipole—dipole attractions London dispersion forces	48

Compound	Class	Molar mass (g mol ⁻¹)	Type of intermolecular force	Boiling point (°C)
Methyl methanoate	Ester	60.1	Dipole—dipole attractions London dispersion forces	32
Chloroethane	Halogenoalkane	64.5	Dipole—dipole attractions London dispersion forces	12
Ethoxymethane	Ether	60.1	Dipole—dipole attractions London dispersion	7.4

Compound	Class	Molar mass (g mol ⁻¹)	Type of intermolecular force	Boiling point (°C)
			forces	
Butane	Alkane	58.1	London dispersion forces	-0.5

From **Table 1**, we can make the following assumptions. Compounds with hydrogen bonding between their molecules have higher boiling points. And compounds with only London dispersion forces have lower boiling points. In terms of melting points, the trend is not as clear as that for the boiling point. The reason for this is that the melting point depends on the ability of the molecules to pack closely together when they form a solid, so molecules that can pack more closely together in the solid form tend to have higher melting points.

Worked example 1

For each pair of compounds, predict which one has the higher boiling, giving a reason in each case.

1. Methanal and ethanal
2. 2-methylpentane and 2,3-dimethylbutane
3. Propane and ethanol

1. Both are aldehydes so have dipole–dipole attractions and London dispersion forces between their molecules.
Ethanal has a higher molar mass so it has stronger London dispersion forces and a higher boiling point.
2. Both molecules are non-polar so have London dispersion forces between their molecules. 2,3-dimethylbutane is more branched so has weaker London dispersion forces between its molecules and a lower boiling point.

3. The molar masses of both compounds are quite similar but propane, being a non-polar molecule, only has relatively weak London dispersion forces between its molecules. Ethanol has hydrogen bonding between its molecules so has a higher boiling point.

Making connections

What is the influence of the carbon chain length, branching and the nature of the functional groups on intermolecular forces? In [section S2.2.9](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-intermolecular-forces-id-43852/>), you learned about the different factors that affect the strength of the intermolecular forces between molecules.

Theory of Knowledge

You have seen how the shape of a molecule can influence the strength of the intermolecular forces, affecting the melting and boiling point. Fats, such as butter and animal fat, contain saturated hydrocarbon chains that are straight and experience stronger London dispersion forces, resulting in a higher melting point and a solid at room temperature. Oils, such as olive or peanut oil, contain unsaturated hydrocarbon chains that are bent at each C=C, resulting in weaker London dispersion forces and lower melting points, making these liquids at room temperature.

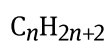
Liquid oils are more prone to spoilage than solid fats and can be converted to fats in a process called hydrogenation. Hydrogenated and partially hydrogenated fats have a longer shelf life, but these substances have been linked to health problems, including heart disease. Several countries have imposed limits or even banned foods containing these fats. Should knowledge of adverse health effects dictate an ethical responsibility?

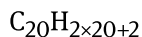
Members of the same homologous series also have the same general formula, which for the alkanes is C_nH_{2n+2} . Using this general formula, the molecular formula of any alkane can be determined if the number of carbon atoms is known.

Worked example 2

Determine the molecular formula of an alkane with 20 carbon atoms given that the general formula is C_nH_{2n+2} .

Using the general formula:





Next, we will look at another homologous series, the alcohols, the first four members of which are shown in **Figure 4**.

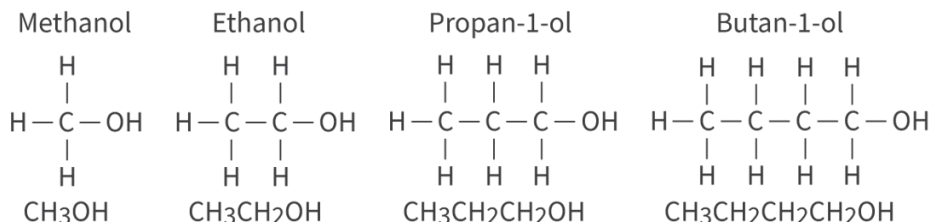


Figure 4. The full structural formulas of the first four members of the alcohol homologous series, methanol to butan-1-ol.

 More information for figure 4

The image displays the structural formulas of the first four members of the alcohol homologous series: methanol, ethanol, propan-1-ol, and butan-1-ol. Each compound is depicted with its respective chemical formula. Methanol is shown as CH₃OH, with a carbon atom bonded to three hydrogen atoms and one hydroxyl group (OH). Ethanol has the formula CH₃CH₂OH, consisting of two carbon atoms, five hydrogen atoms, and a hydroxyl group. Propan-1-ol is represented as CH₃CH₂CH₂OH, with a three-carbon chain and a total of seven hydrogen atoms along with a hydroxyl group. Lastly, butan-1-ol is illustrated as CH₃CH₂CH₂CH₂OH, featuring a four-carbon chain and nine hydrogen atoms, along with a hydroxyl group. The image highlights the sequential increase in carbon and hydrogen atoms as the series progresses, maintaining the hydroxyl group in each structure.

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Think about the following questions before checking the answers below.

1. How do the successive members of the homologous series differ?
2. What is the functional group that is common to all members of the series?
3. Do the members have the same general formula?
4. Would you expect to see a gradation (gradual increase) in the boiling points? If so, how would you explain it?

1. The successive members differ by a $-\text{CH}_2-$ group. The molar mass increases by approximately 14 g mol^{-1} between

- each successive member.
- The common functional group is the hydroxyl group (—OH). All members of the same homologous series have the same functional group.
 - The general formula is the same for all members of the alcohol homologous series.
 - The boiling points show a graduation from 78°C for ethanol to 118°C for butan-1-ol. The increasing molar mass results in stronger London dispersion forces between the molecules, which results in an increased boiling point.

Table 2 shows the general formulas for all the homologous series covered in the DP chemistry course.

Table 2. The general formulas for the homologous series.

Class	Functional group / structural feature	General formula
Alkane	$\text{—CH}_2\text{—CH}_2\text{—}$	$\text{C}_n\text{H}_{2n+2}$
Alkene	—CH=CH—	C_nH_{2n}
Alkyne	$\text{—C}\equiv\text{C—}$	$\text{C}_n\text{H}_{2n-2}$
Halogenoalkanes	R—X	$\text{C}_n\text{H}_{2n+1}\text{X}$
Alcohol	R—OH	$\text{C}_n\text{H}_{2n+2}\text{O}$
Aldehyde	R—CHO	$\text{C}_n\text{H}_{2n}\text{O}$
Ketone	R—CO—R	$\text{C}_n\text{H}_{2n}\text{O}$
Ether	R—O—R	$\text{C}_n\text{H}_{2n+2}\text{O}$
Ester	R—COO—R	$\text{C}_n\text{H}_{2n}\text{O}_2$
Carboxylic acids	R—COOH	$\text{C}_n\text{H}_{2n}\text{O}_2$
Amines	R—NH_2	$\text{C}_n\text{H}_{2n+3}\text{N}$
Amides	R—CONH_2	$\text{C}_n\text{H}_{2n+1}\text{NO}$

In this activity, you will use condensed structural formulas and investigate physical properties for a homologous series.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Complete the following homologous series by filling in the missing condensed structural formulas of the compounds.

(a)

Compound	Homologous Series		
CH ₄		CH ₃ CH ₂ CH ₃	CH ₃

(b)

Compound	Homologous Series		
HCHO			CH ₃ CH ₂ CH

(c)

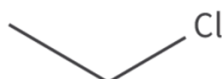
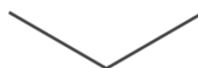
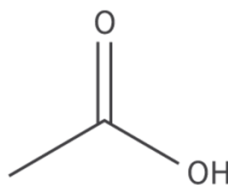
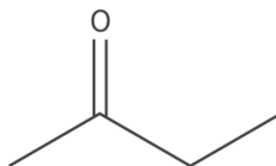
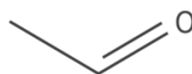
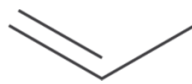
Compound	Homologous Series		
HCOOH			CH ₃ CH ₂ CO ₂ H

(d)

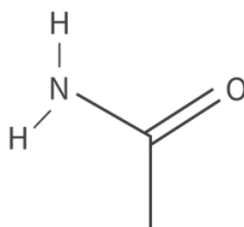
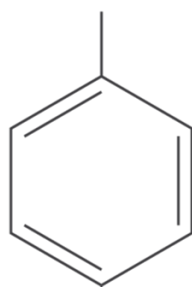
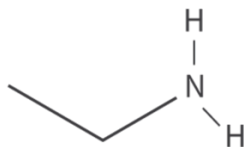
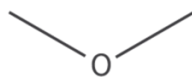
Compound	Homologous Series		
CH ₃ COCH ₃			CH ₃ COCH ₂ CH ₃

1. Identify to which homologous series each of the following molecules belongs.

Skeletal formula



Skeletal formula



Skeletal formula



Choose any one of the homologous series featured above. Use an online database such as [Pubchem](https://pubchem.ncbi.nlm.nih.gov/) (<https://pubchem.ncbi.nlm.nih.gov/>) to find the boiling points or melting points of the first five members of your chosen series. Using Excel or any suitable graphing software, plot a graph of boiling point or melting point against molar mass. Explain the trend using your knowledge of intermolecular forces.

1. Complete the following homologous series by filling in the missing condensed structural formulas of the compounds.

(a) CH_4 , CH_3CH_3 , $\text{CH}_3\text{CH}_2\text{CH}_3$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

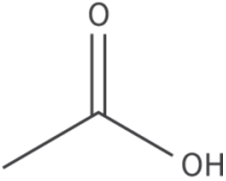
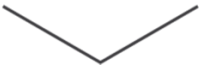
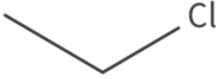
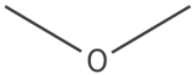
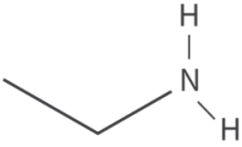
(b) HCHO , CH_3CHO , $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$

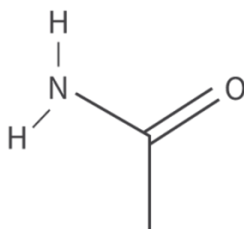
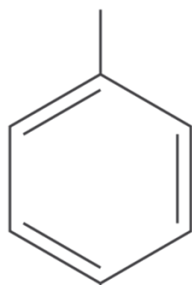
(c) HCO_2H , $\text{CH}_3\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$

(d) CH_3COCH_3 , $\text{CH}_3\text{COCH}_2\text{CH}_3$, $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$,
 $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

2. Identify to which homologous series each of the following molecules belongs.

Skeletal formula	

Skeletal formula	
	
	
	
	
	
	
	
	

Skeletal formula

1. The first five members of any homologous series should show an increase in the melting and boiling points. The explanation of the trend should refer to the increasing strength of the London dispersion forces between the molecules as the molar mass increases.